DATA ANALYTICS · PERSONAL PROJECT

Stock Market Intelligence & Trend Analysis Dashboard

Turns raw stock price data into dashboards showing trends, moving averages, and sector-wise performance.

Built using a practice dataset · Kasthuri Sathish

The problem

Raw stock data is large and hard to read. Investors and analysts need a quick way to see which stocks and sectors are moving, and how prices are trending over time.

What I built

An end-to-end analytics pipeline that cleans and aggregates stock data, calculates technical metrics, and presents them in interactive Power BI and Streamlit dashboards.

How it works

Raw stock price data → Clean and aggregate (Pandas) → Store and query (SQL) → Moving averages and trends → Sector-wise analysis → Power BI and Streamlit dashboards

Key product decisions

- Organized analysis by sector, so users can compare industries, not just single stocks.
- Used moving averages to show the trend behind noisy daily prices.
- Built in both Power BI and Streamlit, for business users and a web-based view.
- Made 50-day and 200-day simple moving averages the main view, so golden cross and death cross signals stand out.

Testing and learnings

- Cross-checked moving averages and daily percentage changes against the raw CSV price records using SQL validation queries.
- Optimized SQL aggregation queries and added Streamlit caching to speed up multi-year charts.
- Clean data matters most: missing or duplicate prices can distort averages and trends.
- Limitation: it shows past performance, not predictions or investment advice.

Tech stack

Python · Pandas · NumPy · SQL · Power BI · Streamlit

GitHub: github.com/kasthuri-tech/Financial-Domain-Stock-Market-Intelligence-and-Trend-Analysis
Portfolio: kasthuri-tech.github.io